

An Analysis of the Impact of Student Life Centers of Interest on Rental Price in Beer Sheva

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Introduction:

This project examines the impact of points of interest (POIs) to the student community in Beer Sheva on rental prices within the primary “student neighborhoods” (Bet, Gimel, Dalet). The broader context of this research addresses how local rental markets are shaped not only by structural apartment characteristics but also by the surrounding urban environment. In student-dense areas, renters typically place a premium on proximity to locations that facilitate daily campus life, such as the university, cafés, restaurants, bars, parks, and central commercial streets. Consequently, apartments located near these hubs may exhibit distinct rental price variations compared to those located further away.

Our specific research question asks: How do centers of interest among the student population affect rent prices in Beer Sheva?

Focusing on the Be'er Sheva rental market, we examine whether walking distance from these POIs is significantly associated with rent, controlling for structural apartment features such as size, room count, neighborhood, balconies, and safe rooms (Mamad). This inquiry is particularly relevant given Be'er Sheva's substantial student population, whose localized demand heavily influences the rental market surrounding Ben-Gurion University.

Previous research provides several foundational frameworks for this study. Fields developed a hedonic pricing model for off-campus student housing across six university markets, demonstrating that distance from campus can correlate with rent in a non-linear fashion. Furthermore, Baron and Kaplan examined the broader impact of “studentification” on private rental housing markets, highlighting how high concentrations of students affect monthly rental prices. Building upon this literature, our approach utilizes both apartment-level characteristics and spatial distance variables to analyze local rental listings. Following an initial descriptive exploration of the data, we deploy both statistical and predictive machine learning models to evaluate the explanatory power of student POIs on rent prices, thereby extending the concept of student-oriented locational value to the Be'er Sheva market.

Data Overview:

The primary dataset comprises rental apartment listings in Be'er Sheva, , filtered explicitly for neighborhoods characterized by high student density (Bet, Gimel, Dalet). The finalized dataset encompasses 299 unique observations and 27 variables. Data were aggregated from multiple online real estate platforms, including Yad2, Komo, Homendy, Facebook, Madlan, Homeless, and OnMap.

The primary unit of observation within the dataset is an individual rental listing. The feature variables pertinent to the research question are categorized into the following domains:

- **Structural Features:** Variables detailing the physical property, including monthly rental price, number of rooms, total square footage, property type, and specific amenities (balcony, safe room/Mamad).
- **Locational Features:** Geographic metadata encompassing the neighborhood, street name, latitude, longitude, and spatial geometry.
- **Spatial POI Features:** Continuous variables representing the approximate walking distance in meters from each property to key student hubs: the nearest Ben-Gurion University gateprominent student social centers (Ringelblum, Soprim Park, and The Block Metzadah Street), as well as the nearest bar, café, and restaurant.
- **Source Features:** Metadata concerning the listing identification, descriptive text, and the platform of origin.

Methods and results

Data preprocessing commenced with the removal of outliers to isolate the specific demographic market of interest. The dataset was restricted to properties with monthly rents ranging between 1,500 NIS and 8,000 NIS. This truncation filtered the relevant student market segment, omitting anomalies such as data-entry errors, single-room sublets, and luxury real estate. To mitigate geographical bias and prevent the overrepresentation of any single zone, the sample size was balanced across the three primary neighborhoods (Bet, Gimel, Dalet).

A logarithmic transformation was subsequently applied to the dependent variable (rental price). The original distribution of rental prices exhibited a pronounced right-skew, indicating that a small number of very expensive apartments could strongly affect the model. Computing $\log(\text{price})$ reduced the influence of extreme prices, stabilized the variance, and allowed the coefficients to be interpreted approximately in terms of relative percentage changes. Similarly, walking distances to student POIs were transformed utilizing $\log(\text{distance} + 1)$, since the effect of distance is expected to be pronounced at shorter distances and weaker at larger distances.

Following exploratory correlation analysis utilizing heatmaps, we fitted an Ordinary Least Squares (OLS) multiple linear regression model, which included apartment characteristics, neighborhood, and all distance variables to the centers of interest. The explanatory variables included neighborhood, number of roommates, apartment spaciousness, balcony, Mamad, and log walking distances to the university gate, Ringelblum, Sophrim Park, Metzadah, The Block center, the nearest bar, the nearest cafe, and the nearest restaurant.

However upon inspection of the model outputs, it became clear that the assumptions required for a linear regression model were not fully satisfied. In addition, a VIF test indicated the presence of a significant multicollinearity issue between the neighborhood variable and some of the distance variables. Upon reflection this result is intuitively reasonable as both neighborhoods and distances to points of interest represent similar geographic information.

To build a cleaner and more interpretable linear regression model, we performed manual stepwise feature selection. We excluded the neighborhood variable along with all highly overlapping distance variables and retained a reduced set of explanatory variables:

- Estimated number of roommates (based on reported number of bedrooms in the advert),
- Apartment spaciousness (calculated as the deviation from the normalized mean of the neighborhood),
- Has Mamad (Boolean value as reported in the advert),
- Has balcony (Boolean value as reported in the advert),
- Log walking distance to the nearest university gate,
- Log walking distance to Sophrim Park,
- Log walking distance to the nearest restaurant,
- Log walking distance to the nearest bar,
- Log walking distance to the Block center.

This reduced set of features solved the multicollinearity issue and proved to allow for all model-assumption tests to be fully satisfied.

Despite passing base linear diagnostics, existing literature indicates that distance from entertainment centers frequently exhibits a non-linear relationship with property valuation. Instead, the relationship may be a sort of spatial tradeoff, as in the case whereby tenants want to be within easy walking distance of a bar, but they generally do not want to live directly above one due to the noise.

In an attempt to address this, we fitted an additional linear regression model that included quadratic terms (X^2). Under the assumption that adding both (X) and (X^2) would allow the model to capture a curved relationship, as in those where the effects of distance on rental price may be different at short and long ranges.

The quadratic model exhibited several issues. The assumption of normally distributed errors was not satisfied. In addition, including both a variable and its squared term created high multicollinearity for the university-gate distance and nearest-bar distance variables. This is expected because (X) and (X^2) are mathematically related.

Subsequently, we fitted a Generalized Additive Model (GAM) in an attempt to capture these non-linear spatial dynamics with greater flexibility. While apartment characteristics remained linear predictors, the distance variables were modeled as smoothed functions. This means that instead of forcing each distance variable to have a straight-line effect on rental price, the GAM allowed the relationship to take a curved shape when supported by the data.

The GAM model satisfied both model assumptions: absence of multicollinearity and sufficient model flexibility, meaning that the smoothing function selected for the nonlinear variables was appropriate.

The model demonstrated robust goodness-of-fit, explaining **78.86%** of the null deviance in log rental prices, yielding a Mean Absolute Error (MAE) of **0.1031** and a Root Mean Square Error (RMSE) of **0.1364** in log-price terms. These results indicate a strong fit to the observed data.

The GAM results indicated that the distance to proximity Park HaSophrim and the nearest bar exerted the most statistically significant non-linear effects on rental price.

The data for Sophrim Park reveals a distinct spatial pricing pattern. While prices directly bordering the park are relatively low, they reach their peak at a short walking distance before experiencing a sharp drop-off further out (at approximately 6.3 log meters) away.

Walking distance to the nearest bar showed a wave-like (sinusoidal) pattern, suggesting a noise-nuisance effect. Prices peak right next to the bar for convenience, drop at medium distances where noise outweighs that convenience, and recover further away in quieter areas.

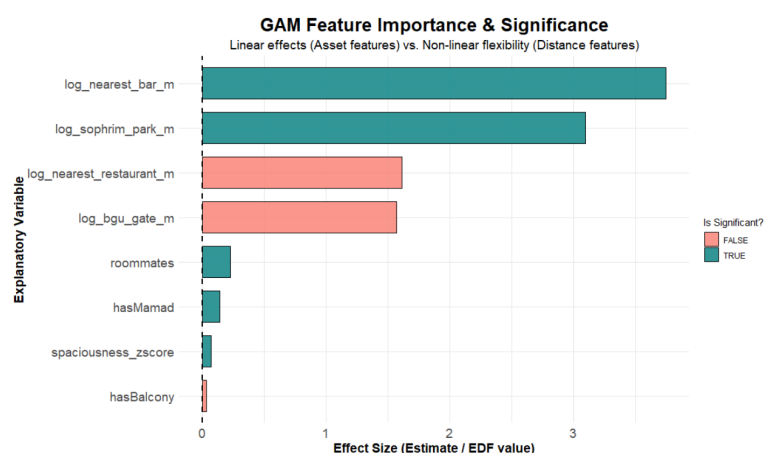
Conversely, the distance to the nearest university gate and the nearest restaurant showed a simple, monotonic linear decline and were not statistically significant when modeled alongside competing covariates. Indicating that the impact of student centers of interest on price is not uniform and changes by location.

Interestingly, while apartment features like roommates, as Mamad, and apartment spaciousness were statistically significant, their overall impact on price fluctuations was lower than that of the main distance variables.

In order to account for complex, multidimensional interactions between spatial variables, two Ensemble-Tree based machine learning regression algorithms were trained: Random Forest and Extreme Gradient Boosting (XGBoost).

The reasoning for our choosing these models was our observation, during the data exploration portion of the research project, of seemingly non-linear relationships between the explanatory distance variables. For example, the effect of distance to the university gate may interact with the distance from the Sophrim park and be mitigated by the side of the road the apartment is located as in the case of Metzadah street where one side is in the Dalet neighborhood and the other in the Be Neighborhood.

Unlike regression models, tree-based models do not require assumptions such as absence of multicollinearity and so we assumed would be better able to fit our data.



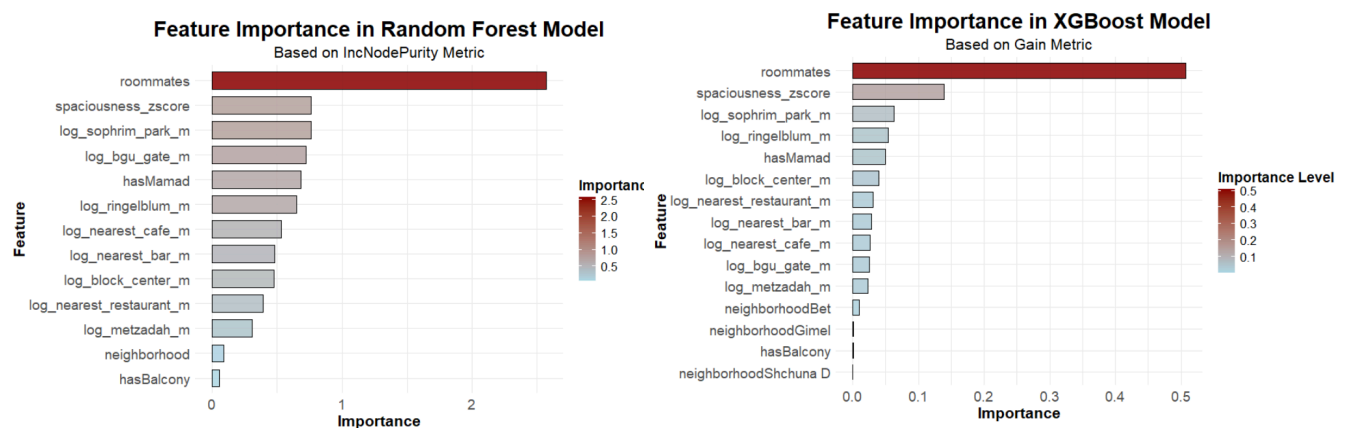
The Random Forest algorithm operates by aggregating a large number of decision trees, each trained on a random subset of the data and features, to produce a final averaged prediction. Conversely, XGBoost utilizes a sequential boosting framework wherein subsequent trees are iteratively fitted to the residuals of preceding trees. While this approach frequently yields superior predictive performance, it inherently necessitates more rigorous hyperparameter tuning.

The Random Forest model yielded robust predictive metrics on the test set, achieving an R^2 of **0.742** indicating that the model accounts for approximately 74.2% of the variance in rental prices. Furthermore, the model produced a Mean Absolute Error (MAE) of 392 NIS, a Root Mean Square Error (RMSE) of 607 NIS, and a Mean Absolute Percentage Error (MAPE) of 11.53%. These figures denote a commendable predictive capacity, particularly given the inherent limitations of the dataset, which omits highly influential unobserved variables such as interior condition, floor level, and furnishing status.

The XGBoost model performed significantly better, achieving an R^2 of 0.848, an MAE of 258 NIS, an RMSE of 440 NIS, and a MAPE of 7.5%.

In other words, XGBoost explained approximately 85% of the variation in rental prices in the test set, with an average relative error of less than 8%.

As such, between the two tree-based models, XGBoost would seem to be the better suited predictive model. The substantial performance margin between the two techniques suggests that the sequential boosting approach was better suited to capture the complex, non-linear pricing dynamics exhibited by the subset local rental market we .



Feature importance analyses of both models confirmed that structural attributes (roommates, apartment spaciousness) served as primary splitting criteria in both, establishing baseline valuation tiers. However, spatial variables—specifically distances to Sophrim Park, Ringelblum, and the university gate—retained substantial predictive importance within those established tiers.

In the Random Forest model, the three most influential distance variables were the walking distance to university gate, Sophrim Park, and Ringelblum.

In the XGBoost model, the three most influential distance variables were Sophrim Park, Ringelblum, and The Block center.

In both models, the neighborhood variables and presence of a balcony had almost no influence. The presence of a Mamad served to have a relatively large effect, which is reasonable in Israel.

The difference in conclusions between the tree-based models and the GAM model stems from the fact that tree-based models identify property size and roommates as primary 'threshold conditions' that dictate the baseline price range, thus placing them at initial splits. In contrast, the GAM model operates in an 'additive' manner—isolating the effect of each variable individually. Because it allows distances to have flexible non-linear relationships, it manages to show that proximity to student centers of interest is actually what best explains the fluctuations and variations in rental price within the market.

In conclusion, student centers of interest significantly yet complexly impact rental prices. While structural property characteristics (spaciousness, number of roommates, etc..) set the baseline price, spatial proximity to academic and social hubs drives subtle, non-linear market fluctuations.

Ultimately, the Be'er Sheva student rental market is profoundly shaped by the spatial accessibility of daily academic and social infrastructure, highlighting the weight that students place on quality of life and the "studential experience" that so draws many to the university.

Limitations and Future Work

A notable limitation of this study is the omission of influential unobserved variables, including interior property condition, floor level, furnishing status, parking availability, renovation history, and specific lease terms. Furthermore the dataset constitutes a cross-sectional snapshot of a specific time period and to apartments aggregated from online platforms, which may affect how well the findings generalize to the entire student rental market in Be'er Sheva.

Additionally, the high geographical density of several student POIs presents significant spatial autocorrelation, complicating the isolation of the independent marginal effect of each hub. This spatial clustering is reflected in the varying coefficient significance across the OLS, GAM, and ensemble tree models.

Given more time, we would expand the dataset by collecting more listings over a longer period and adding missing apartment-level features. With a larger and richer dataset, we could better evaluate the relationship between student centers of interest and rent prices, and examine whether the patterns found in this project remain stable across a broader sample.

Sources:

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